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Failure analysis of overlying strata in fault fracture zone during coal mining

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Abstract

Faults encountered during coal mining can compromise the continuity and integrity of the overburden, resulting in considerable differences in the stress, displacement, and failure fields of the rocks surrounding the fault zone. When a working face is located adjacent to a fault, the fault-disturbed overburden becomes activated and unstable along the fault plane, which could lead to mining disasters. The fault-adjacent overburden morphology during mining was analyzed using a physical model. A mechanical model of the stability of the fault-disturbed overburden was constructed. The criteria for determining the sliding failure of the overburden during mining were defined, from which the critical coal pillar width required to maintain the overburden stability was determined. The results indicate that an inverted trapezoidal block forms in the overburden due to the combined effects of mining and faulting. The morphology of this block is influenced by the coal pillar width, the height of the fractured zone, and the dip angles of fault and coal seam. The block is prone to sliding or rotational failure along the fault plane during mining. As the coal seam and fault dip angles increase, the critical coal pillar width for maintaining overburden stability decreases. Conversely, increasing coal seam thickness increases the critical coal pillar width. The critical width of coal pillar was determined to be 176 m, which was verified through field observations performed in the #3307 working face.

Keywords: fault, overlying strata stability, sliding failure, rotational failure, coal pillar

1. Introduction

Faults encountered during coal mining are geologic structures that disrupt strata, impair their continuities and integrity, and form a zone of crushed rocks within the fault zone. Processes associated with faulting produce considerable differences in the stress, displacement, and failure morphology of the rocks surrounding the fault zone (Chen *et al.* 2018b, 2020a, 2020b, 2023).

When mining near a fault, the combined effects of mining and faulting can create considerable differences in the responses of the rock mass near the fault and the intact rock mass. The fault-disturbed overburden can become activated and failure can occur along the fault plane, potentially leading to severe mining disasters, such as roof collapses (Islam & Shinjo 2009; Liu *et al.* 2017; Lin *et al.* 2019), water inrush (Li *et al.* 2011; Cao *et al.* 2022; Ma *et al.* 2022;

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Zhang *et al.* 2022; Liu *et al.* 2023), rock bursts (Sainoki & Mitri 2015; Lu *et al.* 2018; Chen *et al.* 2018a; Cheng *et al.* 2019; Wei *et al.* 2021), and coal and gas outbursts (Cao *et al.* 2001, 2020; Chen *et al.* 2019). These disasters have been primarily caused by mining-related disturbances of the equilibrium state of the strata near the fault. Therefore, understanding the stability characteristics of fault-disturbed overburden is critical for preventing mining disasters.

Researchers have recently examined the stability characteristics and the activation and failure mechanisms of fault-disturbed overburden, thereby establishing a foundation for safe mining in working faces adjacent to faults. Chen *et al.* (2023) developed an experimental system to simulate fault formation and determined overburden morphologies, failure patterns, and stress field variations in the working face under the effects of mining and faulting. Wang *et al.* (2020, 2021a, 2021b) conducted a physical simulation of overburden movement and deformation in the working face adjacent to a fault, obtained the spatial morphological characteristics of the overburden, and analyzed the variations in the displacement and stress fields. Li *et al.* (2013, 2016) constructed a mechanical model of overburden stability in the working face adjacent to a fault based on the voussoir beam structure model and defined the criteria for determining the overburden failure. Jiao *et al.* (2019) investigated the dynamic mechanical responses of mining-disturbed faults; analyzed the feedback mechanisms between rock pressure in the working face, fault damage, and slip; and determined the major factors influencing fault slip failure. Li *et al.* (2019) constructed a mechanical model of mega-thrust fault activation, determined the activation conditions for thrust faults, and analyzed the effects of various factors (e.g. fault dip angle, fault medium cohesion, internal friction angle of the fault medium, and point of action of horizontal thrust) on fault activation. Yu *et al.* (2019a, 2019b) constructed a mechanical model of overburden slip in a fault zone, determined the fault activation mechanism, and defined the mechanical criteria for mining-disturbed fault slip and its major influencing factors. Han *et al.* (2020) analyzed the variations in the stress field in the fault zone during hanging wall and footwall mining, constructed a mechanical model of fault activation during these processes, and defined the mechanical criteria for determining fault slip failure. Liu *et al.* (2023) have proposed a novel method to calculate the waterproofing coal pillar near the water-conducting fault by using elastic–plastic mechanics theory. While previous studies have focused on mining-induced stress and displacement evolution in fault zones, the criteria for determining mining-induced fault activation, few reports have addressed the stability characteristics and activation criteria of fault-disturbed overburden considering its load-bearing structure.

In this study, a physical simulation experiment was conducted based on the working face #3307 in the Yiqiao Coal Mine (Shandong, China). The spatial morphology of the fault-disturbed overburden was analyzed, a mechanical model of the overburden stability was constructed, the criteria were defined for determining the sliding failure of the fault-disturbed overburden, and the critical coal pillar width was determined to maintain the overburden stability. This study not only enhances our understanding of overburden stability in fault-disturbed areas, but also provides crucial insight for improving coal mining safety protocols and operational efficiency, thereby potentially mitigating the risks of severe mining disasters.

2. Physical simulation of overlying strata characteristics

2.1. *Site description*

Working face #3307 of the Yiqiao Coal Mine (Shandong, China) is 985 m long and 200 m wide. The ground elevation ranges from +42.1 to +44.3 m, and mine elevation ranges from −744.1 to −456 m. The thickness of #3 coal seam is 4.3 m on average, and the inclination of the coal seam is 15° on average. The working face is located east of fault YF20 (a normal fault with a dip angle of 40°–60° and a throw of 120–180 m), west of unmined coal seam, south of fault SF6, and north of fault SF21 and gobs #3303 and #3305, which are shown in Fig. 1a.

Based on the key strata theory (Qian *et al.* 2003), the overburden in working face #3307 comprises five key strata. Key strata 1 (KS 1) is an 8.1-m thick fine-grained sandstone, KS 2 is a 6.8-m thick fine-grained sandstone, KS 3 is a 6.35-m thick medium-grained sandstone, and KS 4 is a 6.6-m thick siltstone. In addition, a primary key stratum (PKS) is included, which is an 11.0-m thick siltstone. The locations of the key strata are shown in Fig. 1b.

2.2. *Experimental scheme*

The accuracy of a physical laboratory simulation based on actual engineering conditions and the ability of the resulting experimental data to reflect overburden movement and deformation in the working face are dependent on the degree of similarity between the physical model and the actual geologic structure (Whittaker & Reddish 1989; Wang *et al.* 2019a, 2019b). The parameters of an experimental model should be designed by similarity theory and the experimental conditions. According to similarity theory (Ghabraie *et al.* 2015a, 2015b), the relevant parameters of the experimental

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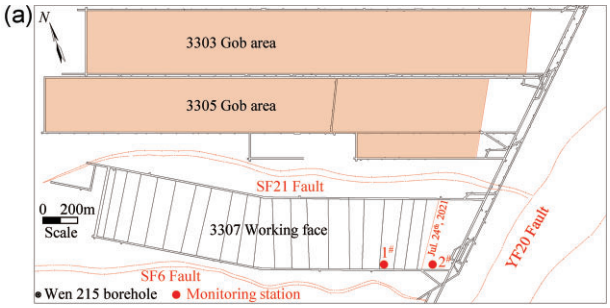

(b)

No.	Thickness (m)	Depth (m)	Lithology	Remarks
29	248.20	248.20	Unconsolidated layers	
28	11.00	259.20	Siltstone	PKS
27	4.50	263.70	Fine grained sandstone	
26	6.20	269.90	Mudstone	
25	1.40	271.30	Siltstone	
24	2.40	273.70	Mudstone	
23	6.60	280.30	Siltstone	KS 4
22	1.40	281.70	Mudstone	
21	3.90	285.60	Fine grained sandstone	
20	4.20	289.80	Mudstone	
19	4.60	294.40	Siltstone	
18	15.30	309.70	Mudstone	
17	3.00	312.70	Siltstone	
16	9.20	321.90	Fine grained sandstone	
15	6.35	328.25	Medium grained sandstone	KS 3
14	1.55	329.80	Fine grained sandstone	
13	3.80	333.60	Mudstone	
12	2.20	335.80	Siltstone	
11	5.90	341.70	Mudstone	
10	6.80	348.50	Fine grained sandstone	KS 2
9	0.30	348.80	Mudstone	
8	0.88	349.68	Mudstone	
7	0.60	350.28	2# Coal seam	
6	0.62	350.90	Mudstone	
5	5.20	356.10	Fine grained sandstone	
4	6.50	362.60	Mudstone	
3	7.00	369.60	Medium grained sandstone	
2	8.10	377.70	Fine grained sandstone	KS 1
1	1.04	378.74	Mudstone	
0	4.30	383.04	3# Coal seam	

Figure 1. Geological conditions at working face #3307. (a) Layout of working face #3307 and (b) stratigraphy determined from borehole Wen 215.

model must fulfill the following equation:

$$\begin{cases} \dfrac{C_\sigma}{C_\rho \times C_L} = 1 \\ C_\sigma = \dfrac{\sigma_P}{\sigma_m}, C_\rho = \dfrac{\rho_P}{\rho_m}, C_L = \dfrac{L_P}{L_m} \end{cases} \tag{1}$$

where C_σ is the strength similarity ratio, C_ρ is the density similarity ratio, C_L is the geometric dimension similarity ratio, P represents the actual geologic structure, and m represents the experimental model.

An experimental platform measuring 200 × 30 × 100 cm (length × width × height) was used in this study. According to similarity theory and the dimensions of the actual geologic structure, the geometric similarity, bulk density similarity, and stress similarity ratios of the model were set as $C_L = 200{:}1$, $C_\rho = 1.6{:}1$, and $C_\sigma = 320{:}1$, respectively. Figure 2 shows the experimental model setup. In the experiment, the coal seam was simulated as 2 cm thick, with a dip angle of 15°, and fault YF20 was modeled with a dip angle of 50° and a throw of 60 cm. The actual coal seam is located at a depth of 600 m. Owing to the experimental platform's size limitations, a compensation load of 46.875 kPa was applied to the top section of the model. A plane stress model was used for the experiment, in which a uniform layer of mica laid between the strata was used to mimic stratification. To avoid strata movement being influenced by boundary conditions, a 15-cm wide coal pillar was reserved on the open-off cut side of the working face. During the experiment, the working face was excavated from right to left at intervals of 5 cm every 15 min.

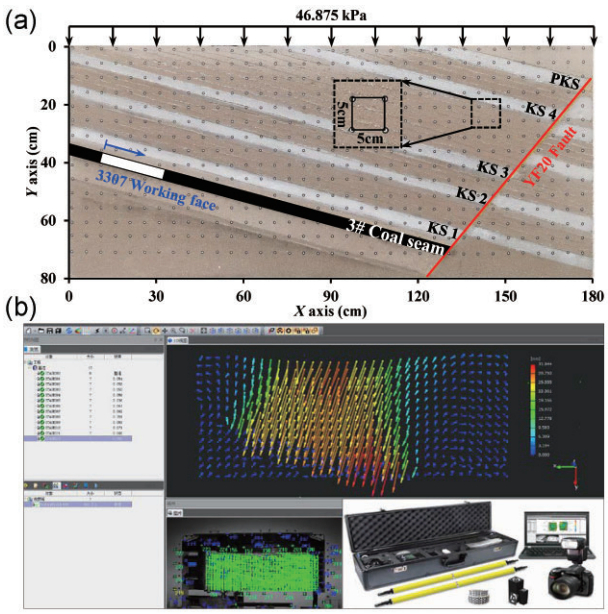

Figure 2. Photos of the (a) experimental model setup and (b) XTDP-DEF three-dimensional photogrammetry system.

The materials used to represent the strata included sand, calcium carbonate, and gypsum. Based on the equivalent material mixing ratios for strata found in the literature (Ghabraie *et al.* 2015a, 2015b; Wang *et al.* 2019a, 2019b), the equivalent material mixing ratios for the various strata in the model were determined (Table 1).

The movement and deformation of the overburden were monitored by using XTDP-DEF three-dimensional photogrammetric system with minimum accuracy of the photogrammetry analysis (±0.015 mm m^{-1}) (Fig. 2b). It was programmed to capture measurements at each stoping step. Figure 2a shows the layout of the displacement measurement points in the experimental model. Specifically, the

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Table 1. Materials used in laboratory tests to represent the bedrock strata.

Rock strata	Thickness (cm)	Material ratio sand:$CaCO_3$: gypsum	Bulk density (kN/m^3)	Elastic modulus (MPa)	Compressive strength (kPa)	Tensile strength (kPa)
Siltstone	5	50:3:7	16.4	131.7	222	30.5
Mudstone	8	50:7:3	16.5	47.9	90	12
Siltstone	3	50:3:7	16.4	131.7	222	30.5
Mudstone	20	50:7:3	16.5	47.9	90	12
Medium-grained sandstone	3	30:7:3	16.5	136.8	229	35.7
Mudstone	8	50:7:3	16.5	47.9	90	12
Fine-grained sandstone	8	70:3:7	16.8	152.3	243	34.6
Mudstone	10	50:7:3	16.5	47.9	90	12
Fine-grained sandstone	8	70:3:7	16.8	152.3	243	34.6
Mudstone	2	50:7:3	16.5	47.9	90	12
#3 coal seam	2	—	14.3	17.9	70	9.7
Mudstone	2	50:7:3	16.5	47.9	90	12
Fine-grained sandstone	8	70:3:7	16.8	152.3	243	34.6
Mudstone	2	50:7:3	16.5	47.9	90	12
Siltstone	5	50:3:7	16.4	131.7	222	30.5
Mudstone	8	50:7:3	16.5	47.9	90	12
Siltstone	4	50:3:7	16.4	131.7	222	30.5
Mudstone	12	50:7:3	16.5	47.9	90	12
Fine-grained sandstone	5	70:3:7	16.8	152.3	243	34.6
Mudstone	8	50:7:3	16.5	47.9	90	12
Medium-grained sandstone	4	30:7:3	16.5	136.8	229	35.7

model surface had 15 horizontal and 36 vertical displacement measurement lines. Measurement points were placed at 5 cm intervals both vertically and horizontally, yielding a total of 540 displacement measurement points.

2.3. *Analysis of experimental results*

Figure 3 shows the movement and deformation characteristics and the displacement vector field variations of the overburden under the combined effects of mining and faulting during stoping.

(i) After a 50-cm advancement in the working face, KS 1 underwent initial fracture and simultaneous fracturing and subsidence of the soft strata it controlled. However, KS 2 did not fracture, and the fractured zone in the working face ended at the lower boundary surface of KS 2 (Fig. 3a). When the working face progressed by 70 cm, KS 2 fractured for the first time, resulting in the simultaneous fracture and subsidence of KS 2 and the soft strata it controlled. KS 3 remained unfractured, and the fractured zone in the working face ended at the lower boundary surface of KS 3 (Fig. 3b). As the working face advanced, KS 3 remained unfractured and was therefore deflected, and the fractured zone ended at the lower boundary surface of KS 3 (Fig. 3c and d).

(ii) During the working face stoping, the overburden fractured periodically after the KS fractured. An inverted trapezoidal block formed between the fracture line of the overburden and the fault plane. The left, right, lower, and upper boundaries of the block were the fracture line of the strata, the fault plane, the upper boundary surface of the coal seam, and the upper boundary surface of the fractured zone, respectively (Fig. 3). As the stoping of the working face proceeded toward the fault, the width of the block decreased in the direction of the upper boundary surface of the coal seam. The height of the block increased markedly as the overburden fractured periodically. The height of the block remained constant when the height of the fractured zone did not change. For example, the block heights were 16, 27, 27, and 27 cm when the working face advanced 50, 70, 90, and 110 cm, respectively.

(iii) During the working face stoping, the overburden above the gob moved, deformed, and even fractured, and the range of overburden movement increased stepwise. When the working face and the fault were widely spaced (e.g. when the working face advanced 50, 70, and 90 cm), the overburden moved near

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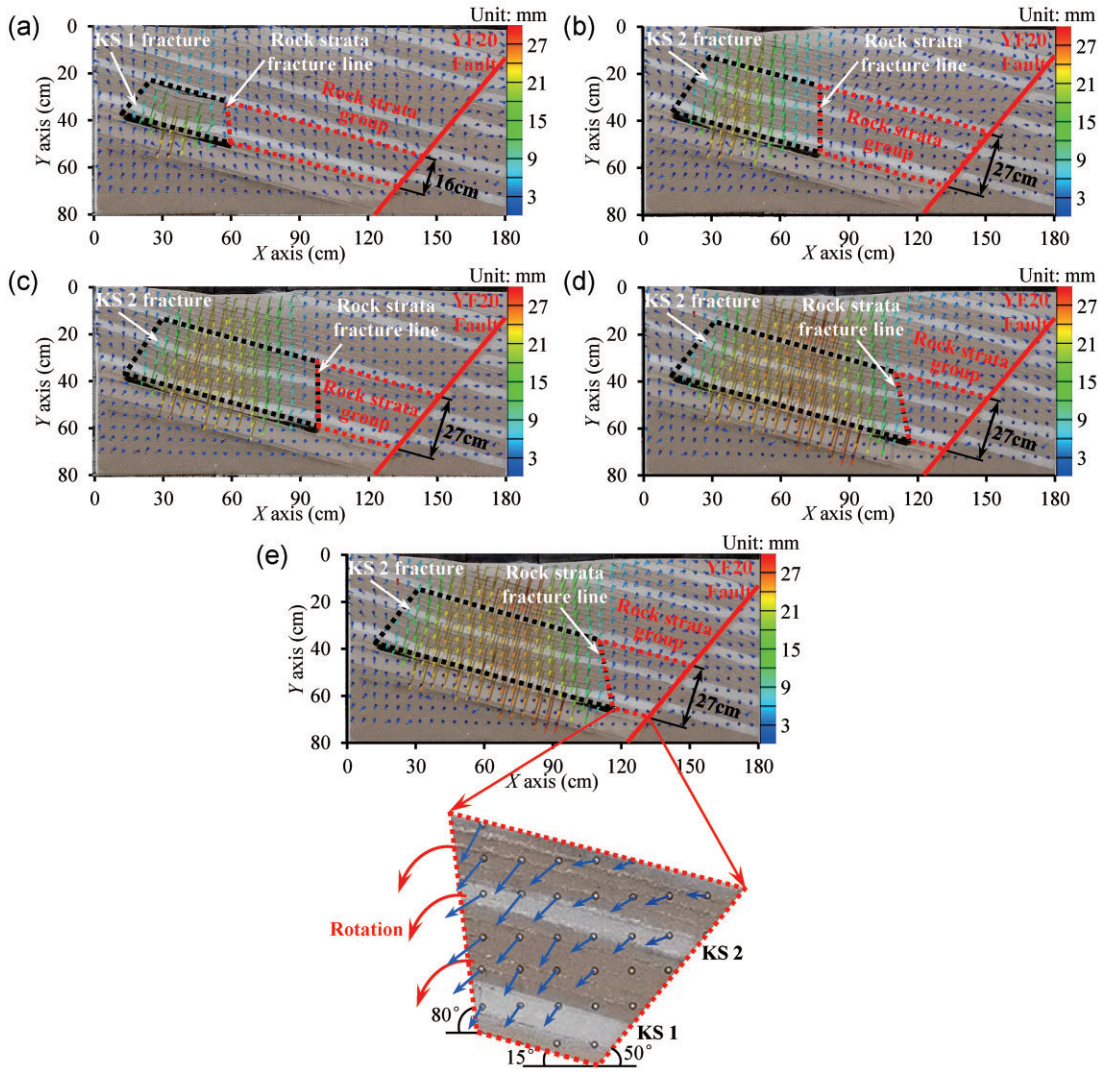

Figure 3. Movement and deformation diagram of overburden after mining in working face. Working face lengths of (a) 50 cm, (b) 70 cm, (c) 90 cm, and (d) 110 cm. (e) Displacement vectors of the inverted trapezoidal block.

the gob, and the block displacement was not considerable. As the working face moved closer to the fault (e.g. when the working face advanced 110 cm), the block displaced and began to rotate toward the gob, as indicated by larger block displacement vectors (Fig. 3e). As the distance between the working face and the fault decreased (i.e. as the width of the coal pillar at the lower boundary of the block decreased), the rotational displacement of the block toward the gob increased further. While undergoing rotational displacement, the block continued to compress the coal pillar. As a result, the coal pillar became susceptible to failure, and the overburden was subjected to sliding failure along the fault plane, which poses a major threat to safe mining adjacent to the fault. Therefore, ensuring safe mining requires a comprehensive block stability analysis.

2.4. *Overlying strata characteristics*

Figure 4 shows the morphology of the overburden after working face #3307 stoping. Based on the KS identification results, the overburden contained four KS and one PKS. KS 1 and 2 fractured sequentially as mining progressed, forming stable voussoir beams #1 and #2, while KS 3 remained intact. The height of the fractured zone ended at lower boundary surface of KS 3. Voussoir beams #1 and #2 in the gob transferred thrust and frictional forces to the unfractured sections of KS 1 and 2 ahead of the working face via hinged joints.

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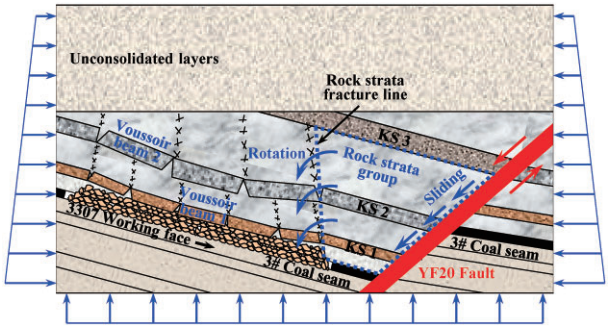

Figure 4. Morphology of the overlying strata in working face #3307.

An inverted trapezoidal block formed between the fracture line of the roof strata and fault YF20. Its left, right, lower, and upper boundaries were the fracture line of the overburden, the fault plane, the upper boundary surface of the coal pillar between the working face and the fault, and the upper boundary surface of the fractured zone, respectively. The block's morphology was influenced by the coal pillar width between the working face and the fault, the height of the fractured zone, and the dip angles of the coal seam and the fault. The height of the fractured zone correlated with the mining height in the working face. Consequently, the block's morphology was influenced by the thickness and dip angle of the coal seam, as well as the dip angle of the fault.

Once the geologic and mining conditions of the working face were established, the block's morphology was primarily influenced by the coal pillar width, which also affected its stability. As the width of the coal pillar decreased, the block may have experienced rotational failure toward the gob and sliding failure along the fault plane. This poses a severe threat to safe mining operations in the working face. Therefore, determining the coal pillar width is crucial for maintaining the stabilities of the block and overburden, thereby ensuring safe mining operations in the working face.

3. Mechanical analysis of the stability characteristics of overlying strata

3.1. *Mechanical model*

Based on the morphology of the overburden during mining in the working face adjacent to a fault, a mechanical model of the block was constructed with a coordinate system (Fig. 5). We assumed that L_1 is the coal pillar width, $q(x)$ is the supporting force from the coal pillar to the block, L_2 is the length of the upper boundary surface of the block, and h is the height of the block. Further, $h_{KS1}, h_{KS2}, \ldots, h_{KSi}, \ldots, h_{KSn}$ symbolize the thickness of KS occurring in the thickness direction of the block; γH is the uniformly distributed load on the upper surface of the block; $T_1, T_2, \ldots, T_i, \ldots, T_n$ are the thrust forces acting from the voussoir beams above the gob on the block (Fig. 5a); $Q_1, Q_2, \ldots, Q_i, \ldots, Q_n$ are the frictional forces between the voussoir beams and the block (Fig. 5b); and $h_{T1}, h_{T2}, \ldots, h_{Ti}, \ldots, h_{Tn}$ indicate the y-direction distances between the points of action of $T_1, T_2, \ldots, T_i, \ldots, T_n$ and the coordinate system origin O, respectively. In addition, σ and τ refer to fault-induced normal and shear stresses on the block, respectively; θ is the dip angle of the coal seam; α is the angle between the fault plane side of the block and the negative direction of the y-axis; and β is the angle between the strata fracture line side of the block and the negative direction of the y-axis.

3.2. *Analytical solutions*

The following equation was obtained based on the static equilibrium in the x and y directions and the moment

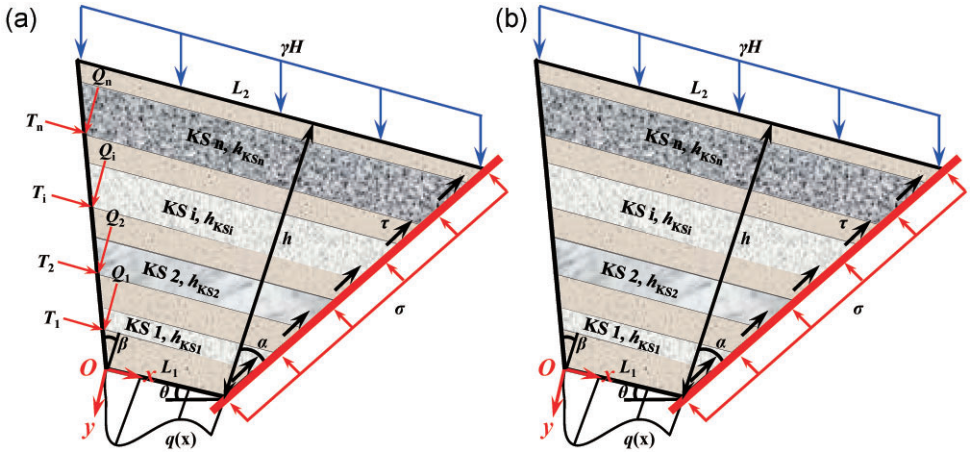

Figure 5. Mechanical model of the rock strata. (a) Stable voussoir beam and (b) voussoir beam failure.

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equilibrium around point O:

$$\begin{cases} \sum F_x = 0 \quad T_1 + T_2 + \cdots + T_i + \cdots + T_n - \sigma h + \tau h \tan\alpha + \gamma H L_2 \sin\theta = 0 \\ \sum F_y = 0 \quad Q_1 + Q_2 + \cdots + Q_i + \cdots + Q_n - \sigma h \tan\alpha - \tau h + \gamma H L_2 \cos\theta - \int_0^{L_1} q(x)\,\mathrm{d}x = 0 \\ \sum M_O = 0 \quad T_1 h_{T1} + T_2 h_{T2} + \cdots + T_i h_{Ti} + \cdots + T_n h_{Tn} - Q_1 h_{T1} \tan\beta - Q_2 h_{T2} \tan\beta - \cdots \\ \quad -Q_i h_{Ti} \tan\beta - \cdots - Q_n h_{Tn} \tan\beta + \gamma H L_2 h \sin\theta + \gamma H L_2 \cos\theta \left(\frac{L_2}{2} - h\tan\beta\right) \\ \quad -\frac{\sigma h}{\cos\alpha}\left(\frac{h}{2\cos\alpha} + L_1 \sin\alpha\right) + \tau h L_1 - \int_0^{L_1} q(x)\,x\mathrm{d}x = 0 \end{cases} \tag{2}$$

Using the geometric characteristics of the block (Fig. 5) and the voussoir beam theory (Qian *et al.* 2003), L_2, (T_1, T_2, T_i, … , T_n) and (Q_1, Q_2, Q_i, … , Q_n) in Equation (2) can be represented as follows:

$$\begin{cases} L_2 = L_1 + h(\tan\alpha + \tan\beta) \\ Q_1 = \dfrac{(4i_1 - 3\sin\varphi_1)\,q_1 l_1}{4i_1 - 2\sin\varphi_1} \qquad T_1 = \dfrac{2q_1 l_1}{2i_1 - \sin\varphi_1} \\ Q_2 = \dfrac{(4i_2 - 3\sin\varphi_2)\,q_2 l_2}{4i_2 - 2\sin\varphi_2} \qquad T_2 = \dfrac{2q_2 l_2}{2i_2 - \sin\varphi_2} \\ \cdots \\ Q_i = \dfrac{(4i_i - 3\sin\varphi_i)\,q_i l_i}{4i_i - 2\sin\varphi_i} \qquad T_i = \dfrac{2q_i l_i}{2i_i - \sin\varphi_i} \\ \cdots \\ Q_n = \dfrac{(4i_n - 3\sin\varphi_n)\,q_n l_n}{4i_n - 2\sin\varphi_n} \qquad T_n = \dfrac{2q_n l_n}{2i_n - \sin\varphi_n} \end{cases} \tag{3}$$

where q_1, l_1, φ_1, and i_1 ($i_1 = h_1/l_1$) correspond to the load, length, rotation angle, and degree of fracture, respectively, of the voussoir beam that formed after the fracture of KS 1 and q_2, l_2, φ_2, and i_2 ($= h_2/l_2$) represent the load, length, angle of rotation, and degree of fracture, respectively, of the voussoir beam that formed after the fracture of KS 2. Similarly, q_i, l_i, φ_i, and i_i represent the load, length, angle of rotation, and degree of fracture, respectively, of the voussoir beam that formed after the fracture of KS i and q_n, l_n, φ_n, and i_n represent the load, length, rotation angle, and degree of fracture, respectively, of the voussoir beam that formed after the fracture of KS n.

The analytical expressions of normal stress σ and shear stress τ on the fault plane during mining in the working face adjacent to a fault can be obtained by combining Equations (2) and (3). These expressions have not been provided here due to space limitations.

If the shear stress τ on the fault plane exceeds the maximum shear stress φ_f owing to mining disturbances, the block slips along the fault plane and rotates toward the gob. Consequently, the potential for the block to slip or rotate along the fault plane can be evaluated by (Li *et al.* 2016)

$$\tan\varphi_f \le \frac{\tau}{\sigma}. \tag{4}$$

Obtaining the friction angle φ_f of a fault plane using field measurements under engineering conditions can be challenging. However, Goodman *et al.* (Li *et al.* 2016). suggested that the friction angle φ_f of a fault plane approximates the internal friction angle of common rocks. Typically, the internal friction angles of sandstone and limestone, as well as those of mudstone and shale, range from 35° to 50° and from 20° to 35°, respectively. Consequently, the friction angle of a fault can be considered 20°–50°, which translates to $\tan\varphi_f = 0.364$–1.192. Moreover, Byerlee's (1978) experimental work indicates that the coefficient of friction between most rocks and a fault plane is $\tan\varphi_f = 0.72$–0.85 when the effective normal stress is <200 MPa. By substituting the analytically derived expressions for normal stress σ and shear stress τ on a fault obtained from theoretical analysis into Equation (4), the width of the coal pillar between the working face and a fault can be determined.

When determining the normal and shear stresses on a fault using theoretical analysis, we assumed that the critical strata fractured to form stable voussoir beams that transferred thrust (T_1, T_2, … , T_i, … , T_n) and frictional (Q_1, Q_2, … , Q_i, … , Q_n) forces to the unfractured sections of the critical strata ahead of the working face. Thus, the coal pillar width can be determined using Equation (4).

It is crucial to note that when the distance between the voussoir beams and the collapsed immediate roof exceeds the critical rotational failure necessary for maintaining the stability of the voussoir beams after the working face stoping, the voussoir beams will slip and become unstable. In this scenario, the voussoir beams no longer transfer thrust and frictional forces to the unfractured sections of the critical strata ahead of the working face; thus, the effects of (T_1, T_2, … , T_i, … , T_n) and (Q_1, Q_2, … , Q_i, … , Q_n) in Equation (4) disappear (Fig. 5b). Despite this, the width of the coal pillar

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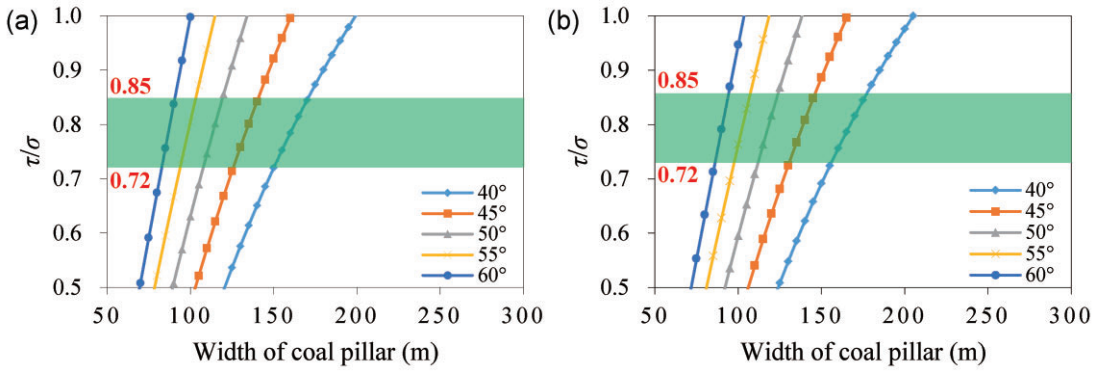

Figure 6. Relationships between τ/σ and coal pillar widths at different fault dip angles when the voussoir beam was (a) stable and (b) failing.

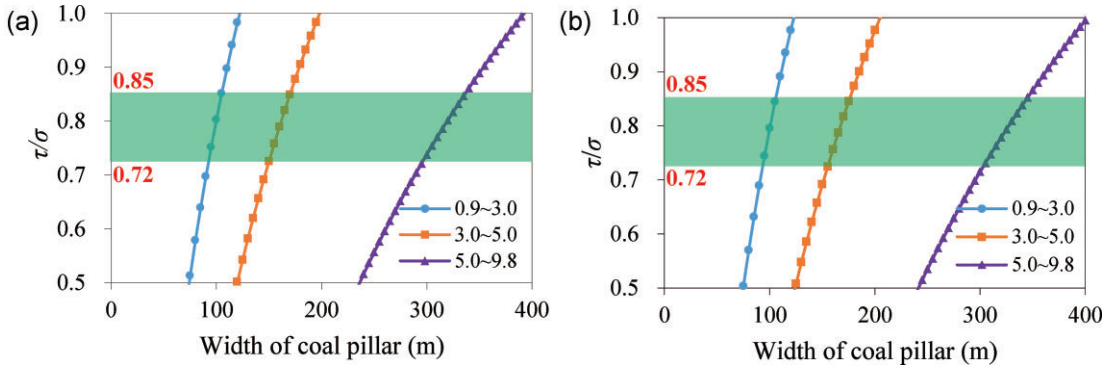

Figure 7. Relationship between τ/σ and coal pillar widths for different coal seam thicknesses when the voussoir beam was (a) stable and (b) failing.

between the working face and fault can still be determined using Equations (2) and (4) when the voussoir beams slip and become unstable. Under actual mining conditions, the width of the coal pillar between the working face and fault should be determined based on a stability analysis of the voussoir beams.

4. Influencing factors of the stability characteristics of overlying strata

Based on the geologic and mining conditions of the working face #3307 and the orientation of fault YF20, the effects of the fault dip angle and coal seam thickness and dip angle on the overburden stability during mining in the adjacent face were analyzed. The parameters used for analysis were $\alpha = 35°$, $\beta = 30°$, $\theta = 15°$, $H = 750$ m, and $\gamma = 25\,000$ kN m^{-3}. The locations and thicknesses of the critical strata in the working face, along with their distances from the coal seam, are shown in Fig. 1b.

Analysis of the morphology of the overburden during mining near the fault indicates that the fractured zone in working face #3307 terminated at the lower boundary surface of KS 3, which implies the presence of two KS (i.e. KS 1 and 2) within the fractured zone. According to the observed rock pressure pattern during stoping in working face #3307, the loads, lengths, and angles of rotation of the voussoir beams that formed after the fracture of KS 1 and 2 were $q_1 = 0.53$ MPa, $l_1 = 12$ m, $\varphi_1 = 5°$ and $q_2 = 0.34$ MPa, $l_2 = 30$ m, $\varphi_2 = 3°$, respectively. By substituting these parameters into equations (2)–(4), the relationships between the width of the coal pillar between the working face and fault and the shear-to-normal stress ratio (τ/σ) on the fault plane were determined for different fault dip angles and various coal seam thicknesses and dip angles. These relationships are shown in Figs 6–8.

4.1. *Fault dip angle*

Figure 6a shows the relationship between the width of the coal pillar and the shear-to-normal stress ratio (τ/σ) on fault YF20 when the voussoir beams were not slipping or experiencing rotational failure. Specifically, when the τ/σ ratio was <0.72, the block underwent sliding along the fault plane, the τ/σ ratio increased, and the width of the coal pillar decreased as the dip angle of the fault increased. When the τ/σ ratio ranged from 0.72 to 0.85, the block was in a critical state between shear slip and stability. For dip angles of 40°, 45°, 50°, 55°, and 60°, the critical minimum widths of the coal pillar were 151, 126, 108, 94, and 83 m, respectively. The critical maximum widths of the coal pillar for the same dip angles

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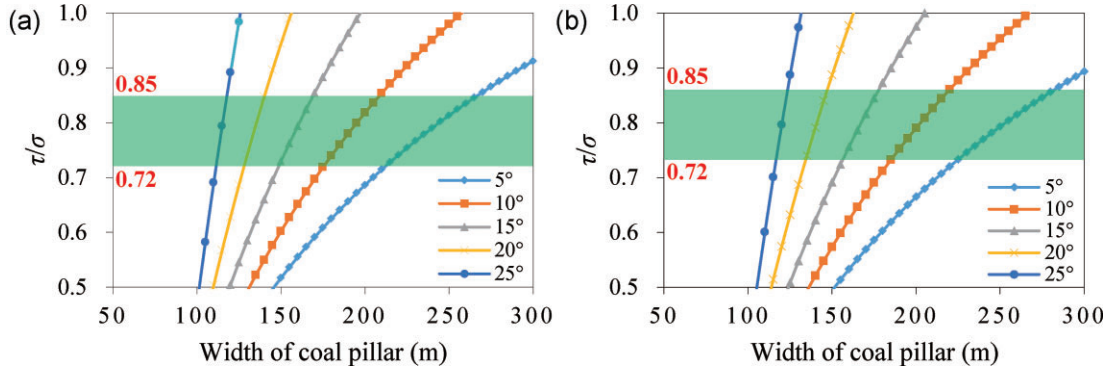

Figure 8. Relationship between τ/σ and coal pillar widths at different coal seam dip angles when the voussoir beam was (a) stable and (b) failing.

were 171, 141, 119, 103, and 91 m, respectively. When the τ/σ ratio exceeded 0.85, the block did not experience sliding along the fault plane.

Figure 6b shows the variations in the coal pillar width concerning τ/σ on fault YF20 when the voussoir beams were slipping or undergoing rotational failure. This pattern is similar to the scenario in which the voussoir beams were not slipping or rotationally deforming; however, the coal pillars were wider when slipping or rotational failure occurred. For example, at a τ/σ ratio of 0.72, the coal pillar widths were 154, 129, 112, 97, and 86 m, which were 3, 3, 4, 3, and 3 m larger, respectively, compared with those in a scenario without slipping or rotational failure. At a τ/σ ratio of 0.85, the coal pillar widths were 176, 145, 123, 107, and 94 m, which were 5, 4, 4, 4, and 3 m larger, respectively, compared with those in a scenario without slipping or rotational failure. When the voussoir beams did not slip or undergo rotational failure, the force acting on the left side of the block disappeared, and the stability of the block could only be maintained by a wider coal pillar.

4.2. *Coal seam thickness*

Since the method for predicting the height of the fractured zone is based on the KS locations, the coal seam thickness and KS locations affect the height of the fractured zone. In turn, they affect the height of the block and the number of KS it contains. According to the KS identification in the Wen 215 borehole (Fig. 1b), when the coal seam thickness ranged from 0.9 to 3.0 m, the heights of the fractured zone and the block both stopped at the lower boundary surface of KS 2 and the block contained only KS 1. When the coal seam thickness ranged from 3.0 to 5.0 m, the heights of the fractured zone and the block stopped at the lower boundary surface of KS 3, which contained KS 1 and 2. When the coal seam thickness ranged from 5.0 to 9.8 m, the heights of the fractured zone and the block stopped at the lower boundary surface of KS 4, and the block contained KS 1, 2, and 3. Incorporating the block heights and the strata thicknesses of the Wen 215 borehole into Equations (2)–(4), the relationship between the width of the coal pillar between the working face and fault and the shear-to-normal stress ratio (τ/σ) on the fault plane was determined for different coal seam thicknesses (Fig. 7).

Figure 7a shows the coal pillar widths concerning τ/σ on fault YF20 when the voussoir beams did not slip or underwent rotational failure. For τ/σ ratios <0.72, the block underwent sliding along the fault plane, resulting in a decrease in τ/σ and an increase in the coal pillar width with increasing coal seam thicknesses. When the τ/σ ratio ranged from 0.72 to 0.85, the block was in a critical state between shear slip and stability. For coal seam thicknesses of 0.9–3.0, 3.0–5.0, and 5.0–9.8 m, the critical minimum coal pillar widths were 91, 148, and 295 m, respectively. The critical maximum coal pillar widths for the same coal seam thicknesses were 104, 169, and 336 m, respectively. When the τ/σ ratio exceeded 0.85, the block did not undergo sliding along the fault plane.

Figure 7b shows the coal pillar widths concerning τ/σ on fault YF20 when the voussoir beams were slipping or undergoing rotational failure. This pattern is similar to the scenario in which the voussoir beams were not slipping or undergoing rotational failure; however, the coal pillars were wider when slipping or rotational failure occurred. For example, at a τ/σ ratio of 0.72, the coal pillar widths were 93, 155, and 304 m, 4, 2, and 9 m larger, respectively, compared with those without slip or rotational failure. At a τ/σ ratio of 0.85, the coal pillar widths were 106, 176, and 345 m, respectively, which were 2, 7, and 9 m larger, respectively, compared with those in a scenario without slip or rotational failure. When the voussoir beams did not slip or undergo rotational failure, the force acting on the left side of the block disappeared, and the stability of the block could only be maintained by a wider coal pillar.

4.3. *Coal seam dip angle*

Figure 8a shows the variations in the coal pillar width related to τ/σ on fault YF20 when the voussoir beams did not slip or were not rotationally deformed. When the τ/σ ratio was <0.72, the block underwent sliding along the fault plane, and

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Table 2. Coal pillar widths influenced by fault YF20 during working face #3307 mining.

Influencing factor		Coal pillar width (m)			
		Stable voussoir beam		Voussoir beam failure	
		$\tan\varphi_f = 0.72$	$\tan\varphi_f = 0.85$	$\tan\varphi_f = 0.72$	$\tan\varphi_f = 0.85$
Fault dip angle (°)	40	151	171	154	176
	45	126	141	130	145
	50	107	119	111	124
	55	94	104	96	106
	60	83	91	85	94
Coal seam thickness (m)	0.9–3.0	91	105	92	106
	3.0–5.0	149	169	154	176
	5.0–9.8	294	335	301	345
Coal seam dip angle (°)	5	212	267	220	276
	10	175	209	182	216
	15	149	169	154	176
	20	128	140	133	146
	25	111	117	116	123

the coal pillar width increased as the coal seam dip angle decreased. When the τ/σ ratio ranged from 0.72 to 0.85, the block was in a critical state between shear slip and stability. For coal seam dip angles of 5°, 10°, 15°, 20°, and 25°, the critical minimum coal pillar widths were 214, 177, 149, 129, and 112 m, respectively, while the critical maximum widths were 262, 209, 169, 140, and 118 m, respectively. When the τ/σ ratio exceeded 0.85, the block did not experience sliding along the fault plane.

Figure 8b shows the variations in the coal pillar width concerning τ/σ on fault YF20 when the voussoir beams slipped or were rotationally deformed. This pattern was identical to the previous scenario in which the voussoir beams did not slip or were not rotationally deformed but with wider coal pillars. For example, at a τ/σ ratio of 0.72, the coal pillar widths were 221, 183, 154, 134, and 117 m, which were 7, 6, 5, 5, and 5 m larger compared with those in a scenario without slip or rotational failure, respectively. Similarly, at a τ/σ ratio of 0.85, the coal pillar widths were 272, 216, 176, 142, and 123 m, which were 10, 7, 7, 2, and 5 m larger than those in a scenario without slip or rotational failure, respectively. When the voussoir beams slipped or underwent rotational failure, the force exerted on the left side of the block disappeared, and maintaining the stability of the block required a wider coal pillar.

5. Engineering applications

5.1. *Coal pillar width determination*

Important factors and corresponding coal pillar widths were determined by examining the effects of the fault dip angle, coal seam thickness, and coal seam dip angle on the overburden stability in the working face adjacent to a fault (Table 2). Based on the specific geologic and mining conditions of the working face #3307 when it approached fault YF20, the maximum required coal pillar width was determined to be 176 m. This width was applicable when the fault dip angle ranged from 40° to 60°, the coal seam thickness was 4.3 m, or the coal seam dip angle was 15°. Therefore, it is essential to maintain a coal pillar width of at least 176 m during mining operations at working face #3307 near fault YF20 to prevent the sliding of the overburden along the fault plane.

5.2. *Field trial results and analysis*

To analyze the rock pressure patterns at different coal pillar widths during working face #3307 mining adjacent to fault YF20, the front abutment pressure was monitored using a KJ649 system, and the working resistances of the supports were monitored using YHY60 pressure transducers. Figure 1 shows the two monitoring stations for the front abutment pressure. Specifically, the distances between stations #1 and #2 and fault YF20 were 300 and 170 m, respectively. Each station had two measurement points at 6 and 12 m depths, spaced 2 m apart. In addition, the YHY60 pressure transducers were installed on specific supports (supports #7, #14, #21, #28, #35, #42, #49, #56, #63, #70, #77, #84, #91, #98, #105, #112, #119, and #126) in the working face to monitor the working resistances.

Stoping in working face #3307 occurred from November 2019 to 24 July 2021, with a coal pillar width of 176 m between the terminal line of the working face and fault YF20. Figure 9 shows the variations in front abutment pressure at stations #1 and #2. At station #1, the front

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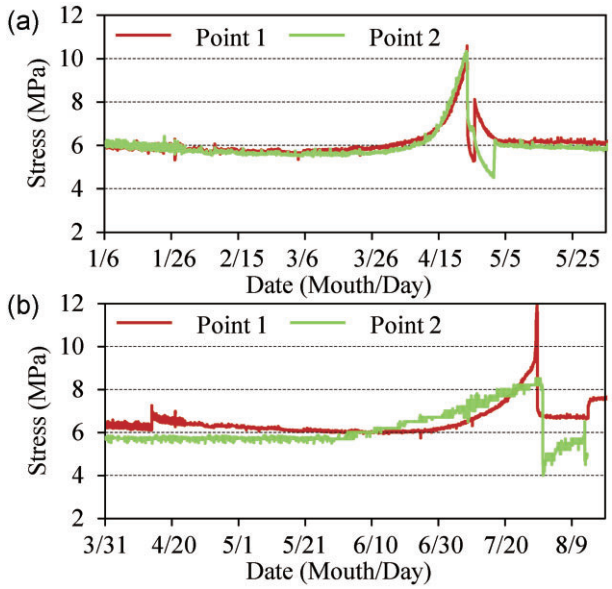

Figure 9. Variations in advanced abutment pressure at stations (a) #1 and (b) #2.

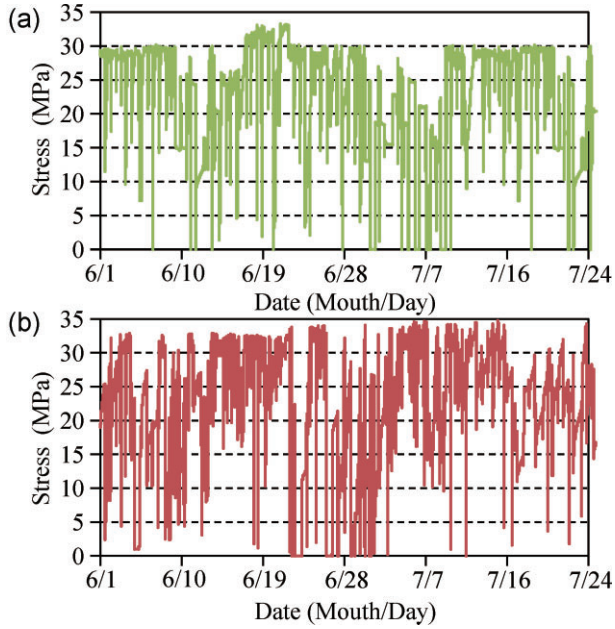

Figure 10. Working resistances at supports (a) #7 and (b) #63 in working face #3307.

abutment pressure began to increase on 6 March 2021, due to mining disturbances in the working face. The pressure reached a maximum on 22 April 2021, and subsequently decreased considerably. The mining disturbance lasted for 60 days. At station #2, the front abutment pressure started to increase on 10 June 2021, owing to mining disturbances, reached a maximum on 27 July 2021, and decreased rapidly afterward. The mining disturbance at station #2 also lasted for 60 days.

As the distance between working face #3307 and fault YF20 decreased from 300 to 176 m, the variations in front abutment pressure and the mining disturbance duration observed at station #1 were identical to those observed at station #2. This suggests that the overburden stability in the working face remained consistent with narrower coal pillar widths, and no sliding of the block occurred along the fault plane.

Figure 10 shows the variations in the working resistances of supports #7 and #63 in the working face, which had distances of 246 and 176 m between the working face and fault YF20 on 1 June and 24 July 2021, respectively. From 1 June to 24 July, the maximum and average working resistances at the end of support #7 were 32.5 and 29.8 MPa, respectively, whereas those at the end of support #63 were 34.8 and 30.3 MPa, respectively. As the width of the coal pillar decreased, the working resistances at the ends of support #7 and #63 exhibited identical variation patterns and similar distribution ranges. This indicates that the overburden stability in the working face remained consistent for narrower coal pillars, and sliding of the block did not occur along the fault plane.

As the coal pillar width between working face #3307 and fault YF20 decreased to 176 m, the variations in the working resistances of the supports and those in the front abutment pressure were identical. This indicates that the fracture morphology and stability of the overburden in the working face did not change considerably. Therefore, the stability of the block was not affected by the narrower coal pillar, and no sliding occurred along the fault plane. This further supports the accuracy and robustness of determining the coal pillar width using block stability analysis.

6. Conclusions

(i) An inverted trapezoidal block formed between the fracture line of the roof strata and fault. The block's morphology is influenced by the coal pillar width, the height of the fractured zone, and the dip angles of the coal seam and the fault. The block will likely undergo sliding failure along the fault plane and rotational failure toward the gob during mining.

(ii) A mechanical model of fault-disturbed overburden stability was constructed to analyze the overburden stability, determine the normal and shear stresses on the block along the fault plane, define the criteria for determining the sliding failure of the fault-disturbed overburden, and calculate the critical coal pillar width required to maintain the overburden stability.

(iii) As the dip angles of the coal seam and fault increased, the normal-to-shear stress ratio on the fault plane increased while the critical coal pillar width decreased. Conversely, an increase in coal seam thickness resulted in a lower normal-to-shear stress ratio and an increase in the critical coal pillar width. The critical coal pillar width increases when the voussoir beam failure.

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(iv) The critical coal pillar width required to maintain the overburden stability in working face #3307 was 176 m. The evolution of working resistances of the supports and front abutment pressure suggests that the coal pillar width determined herein was accurate and reasonable.

(v) Owing to the lack of curves to describe the loading on the upper surface of the inverted trapezoidal block and the supporting force from the coal pillar to the inverted trapezoidal block, the loading and supporting force have been expressed using a linear curve in the mechanical model. Therefore, the mechanical model should be optimized to ensure that the coal pillar width is reasonably determined.

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Conflict of interest statement. The authors declare that they have no conflict of interest.

Data availability

The data that support the findings of this study are available from the corresponding author upon request.

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